As the query shows, the `location` field has stringified latitude and longitude values separated by a comma: `"51.07, 0.08"`. Providing the coordinates in this string format is not the only way to set the `location` field; we can use several other formats to input the `location` geographic coordinates, such as an array, a well-known-text (WKT) point, or a geohash.

Listing 12.3 Indexing in various formats for the `geo_point` data type

```
POST bus_stops/_doc
{
  "text": "London Victoria Station",
  "location" : "POINT (51.49 0.14)"
}

POST bus_stops/_doc
{
  "text": "Leicester Square Station",
  "location" : {
    "lon":-0.12,
    "lat":51.50
  }
}

POST bus_stops/_doc
{
  "text": "Westminster Station",
  "location" : [51.54, 0.23]
}

POST bus_stops/_doc
{
  "text": "Hyde Park Station",
  "location" : "gcpvh2bg7sff"
}
```

Inputs the geo_point as a WKT point (lat, lon)

Inputs the geo_point as a location object

Inputs the geo_point as an array (lon, lat)

Inputs the geo_point as a geohash

These queries index various bus stop locations using multiple formats. We can use a string with the latitude and longitude, as in listing 12.2, or an object, an array, a geohash, or a WKT-formatted `POINT` shape, as in listing 12.3.

Now it's time to learn about the `geo_shape` data type. As the name indicates, the `geo_shape` type helps index and search data using a particular shape: for example, a polygon. Let's see how we can index data for geoshapes.

12.2.2 The geo_shape data type

Like the `geo_point` type, which represents a point on the map, Elasticsearch provides a `geo_shape` data type to represent shapes such as points, multipoints, lines, and polygons. The shapes are represented by an open standard called GeoJSON (http://geojson.org) and are written in JSON format. The geometric shapes are mapped to the `geo_shape` data type.